

Definitions from ComplexAnalysis.org — Section 4.1

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Definition: Complex Sequence

A *complex sequence* is a function whose domain is the positive integers and whose range is a subset of the complex numbers.

Definition 4.1.1: Limit of a Sequence

We say

$$\lim_{n \rightarrow \infty} z_n = \zeta$$

if for every real number $\varepsilon > 0$ there exists an integer N_ε such that

$$|z_n - \zeta| < \varepsilon \quad \text{whenever } n > N_\varepsilon.$$

Definition 4.1.7: Bounded Sequence

A complex sequence $\{z_n\}$ is *bounded* if there exists a real $R > 0$ and integer N such that

$$|z_n| < R \quad \text{for all } n > N.$$

Definition 4.1.9: Cauchy Sequence

A sequence $\{z_n\}$ is a *Cauchy sequence* if for every $\varepsilon > 0$ there exists an integer N_ε such that

$$|z_n - z_m| < \varepsilon \quad \text{whenever } n, m > N_\varepsilon.$$

Definition 4.1.11: Infinite Series

The expression

$$\sum_{k=1}^{\infty} z_k = z_1 + z_2 + \cdots$$

is called an *infinite series*, and the z_k are called the *terms* of the series.

Definition 4.1.12: Convergent Series

If the sequence of partial sums

$$S_n = \sum_{k=1}^n z_k$$

has a limit S as $n \rightarrow \infty$, we say that $\sum_{k=1}^{\infty} z_k$ *converges*, and its sum is S .

Definition 4.1.13: Absolutely Convergent Series

A series $\sum_{k=1}^{\infty} z_k$ is *absolutely convergent* if

$$\sum_{k=1}^{\infty} |z_k|$$

converges as a (real) series.

Definition 4.1.14: Divergent Series

A series that does not converge is said to be *divergent*.

Definition 4.1.22: Cauchy Product

Given two convergent series $\sum_{n=0}^{\infty} a_n$ and $\sum_{n=0}^{\infty} b_n$ with complex terms, the *Cauchy product* is

$$\sum_{n=0}^{\infty} c_n, \quad \text{where } c_n = \sum_{k=0}^n a_k b_{n-k}.$$

Summary

- Complex sequences and their limits
- Boundedness and Cauchy property
- Infinite series: convergence and absolute convergence
- Divergence and products of series

Application of the Cauchy Product: $e^z e^w = e^{z+w}$

Power series definitions:

$$e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!}, \quad e^w = \sum_{n=0}^{\infty} \frac{w^n}{n!}$$

Cauchy product:

Application of the Cauchy Product: $e^z e^w = e^{z+w}$

Power series definitions:

$$e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!}, \quad e^w = \sum_{n=0}^{\infty} \frac{w^n}{n!}$$

Cauchy product:

$$e^z e^w = \sum_{k=0}^{\infty} \sum_{j=0}^k \frac{z^j}{j!} \frac{w^{k-j}}{(k-j)!}$$

Recognize the structure:

Application of the Cauchy Product: $e^z e^w = e^{z+w}$

Power series definitions:

$$e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!}, \quad e^w = \sum_{n=0}^{\infty} \frac{w^n}{n!}$$

Cauchy product:

$$e^z e^w = \sum_{k=0}^{\infty} \sum_{j=0}^k \frac{z^j}{j!} \frac{w^{k-j}}{(k-j)!}$$

Recognize the structure:

$$\frac{1}{j!(k-j)!} = \frac{1}{k!} \binom{k}{j}$$

Application of the Cauchy Product: $e^z e^w = e^{z+w}$

Power series definitions:

$$e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!}, \quad e^w = \sum_{n=0}^{\infty} \frac{w^n}{n!}$$

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$$e^z e^w = \sum_{k=0}^{\infty} \sum_{j=0}^k \frac{z^j}{j!} \frac{w^{k-j}}{(k-j)!}$$

Recognize the structure:

$$\frac{1}{j!(k-j)!} = \frac{1}{k!} \binom{k}{j}$$
$$\Rightarrow e^z e^w = \sum_{k=0}^{\infty} \frac{1}{k!} \sum_{j=0}^k \binom{k}{j} z^j w^{k-j}$$

Binomial theorem:

Application of the Cauchy Product: $e^z e^w = e^{z+w}$

Power series definitions:

$$e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!}, \quad e^w = \sum_{n=0}^{\infty} \frac{w^n}{n!}$$

Cauchy product:

$$e^z e^w = \sum_{k=0}^{\infty} \sum_{j=0}^k \frac{z^j}{j!} \frac{w^{k-j}}{(k-j)!}$$

Recognize the structure:

$$\begin{aligned} \frac{1}{j!(k-j)!} &= \frac{1}{k!} \binom{k}{j} \\ \Rightarrow e^z e^w &= \sum_{k=0}^{\infty} \frac{1}{k!} \sum_{j=0}^k \binom{k}{j} z^j w^{k-j} \end{aligned}$$

Binomial theorem:

$$\sum_{j=0}^k \binom{k}{j} z^j w^{k-j} = (z + w)^k$$

Application of the Cauchy Product: $e^z e^w = e^{z+w}$

Power series definitions:

$$e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!}, \quad e^w = \sum_{n=0}^{\infty} \frac{w^n}{n!}$$

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Generating Function for Bessel Functions J_n

Generating function:

$$\exp\left(\frac{x}{2}\left(t - \frac{1}{t}\right)\right) = \sum_{n=-\infty}^{\infty} J_n(x) t^n, \quad t \in \mathbb{C} \setminus \{0\}$$

Interpretation:

- $J_n(x)$ is the coefficient of t^n
- Encodes all integer-order Bessel functions at once

Immediate consequences:

- Symmetry:

$$J_{-n}(x) = (-1)^n J_n(x)$$

- Differentiation in t produces recurrence relations

Connection:

- Fourier series of a plane wave in polar coordinates
- Separation of variables for the Laplacian in $\mathbb{R}^2$

Recurrence Relations from the Generating Function

Generating function:

$$\exp\left(\frac{x}{2}\left(t - \frac{1}{t}\right)\right) = \sum_{n=-\infty}^{\infty} J_n(x) t^n$$

Differentiate with respect to t :

$$\frac{\partial}{\partial t} \exp\left(\frac{x}{2}\left(t - \frac{1}{t}\right)\right) = \frac{x}{2}\left(1 + \frac{1}{t^2}\right) \exp\left(\frac{x}{2}\left(t - \frac{1}{t}\right)\right)$$

Series side:

$$\sum_{n=-\infty}^{\infty} n J_n(x) t^{n-1}$$

Equate coefficients of t^{n-1} :

$$n J_n(x) = \frac{x}{2} (J_{n-1}(x) + J_{n+1}(x))$$

Three-term recurrence:

$$J_{n-1}(x) + J_{n+1}(x) = \frac{2n}{x} J_n(x)$$